Numerically accurate formulation of implicit turbulent bottom stress in an ocean model with barotropic-baroclinic mode splitting

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Foras na Mara
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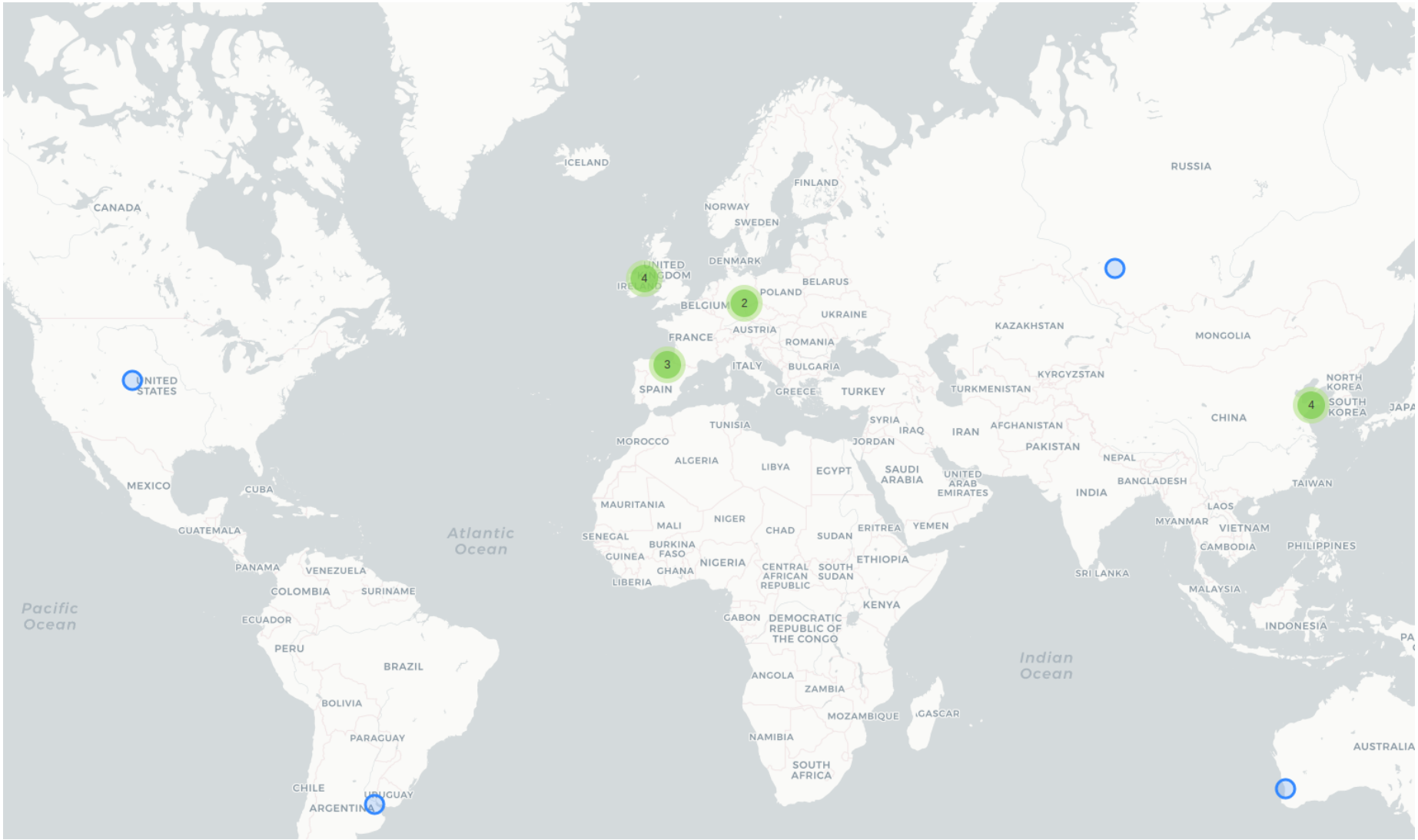
The overarching problem

Bottom drag plays an important role in dissipating tides, and becomes one of the dominant forces in tidal bays and estuaries. One of such places is Galway Bay, Ireland, where tidally induced currents can reach the speed of 2 m/sec, posing challenges in hydrodynamic modeling, essentially due to the interference of different algorithms, which need to work in concert, but originally were not thought to be this way.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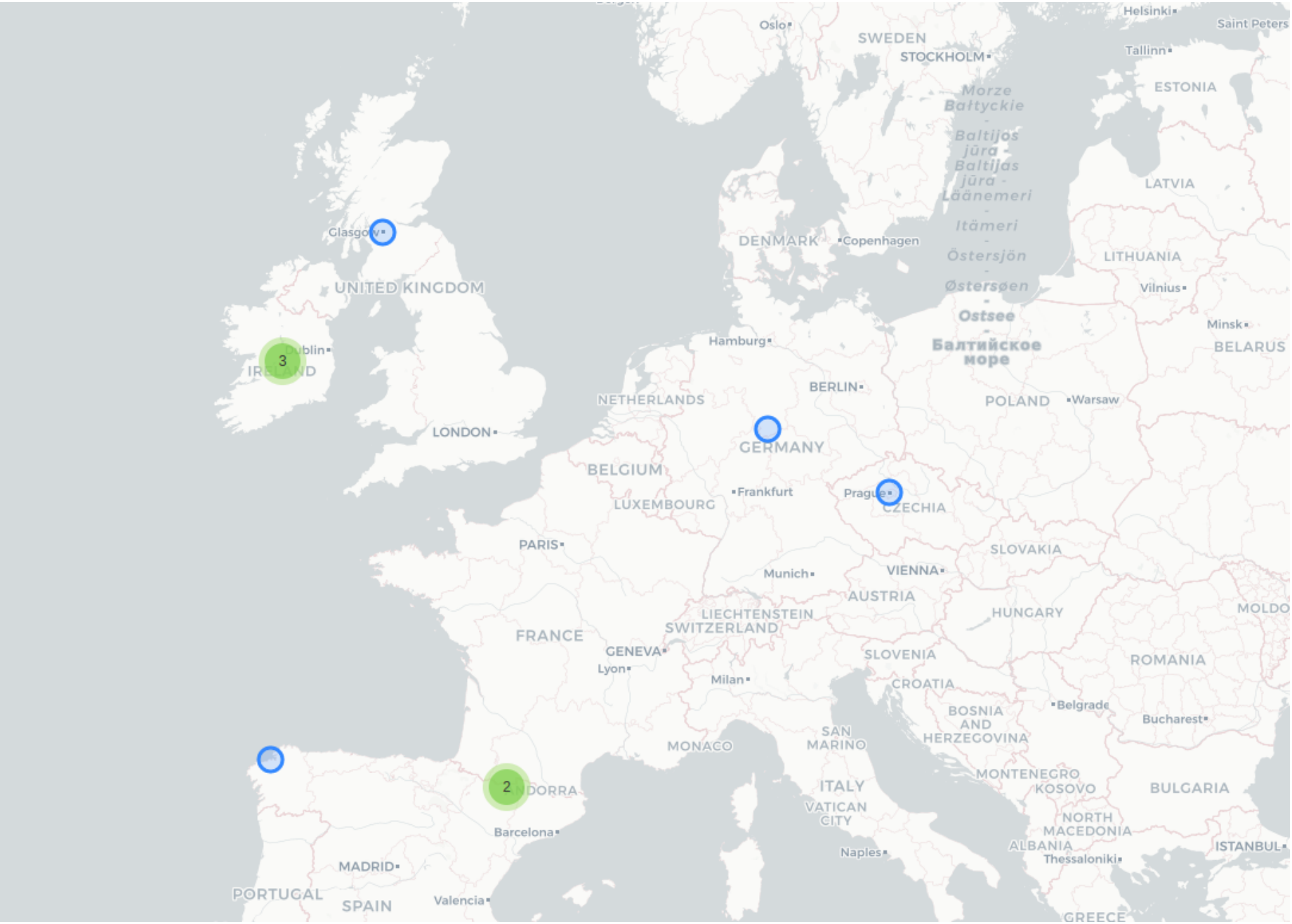
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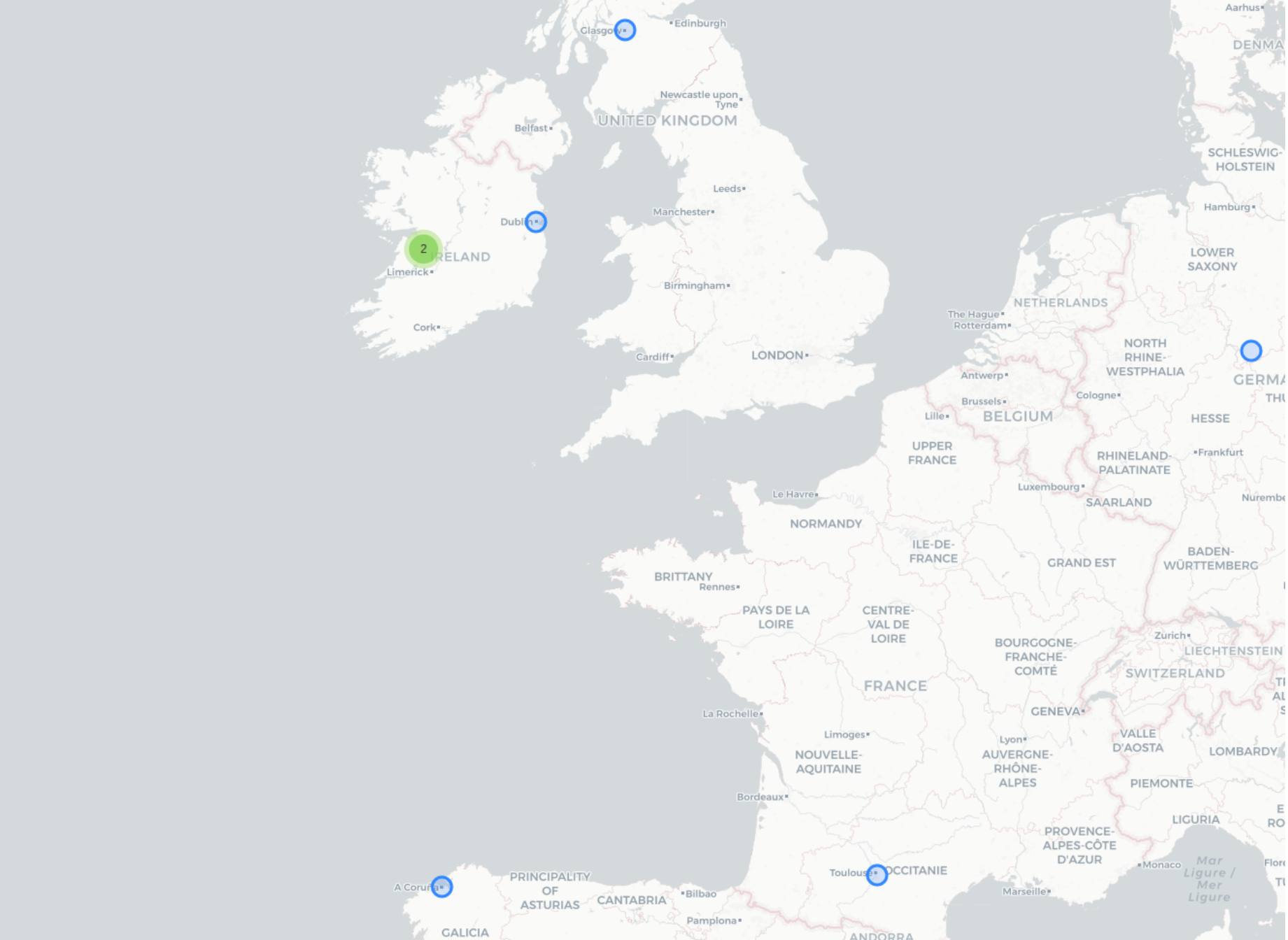
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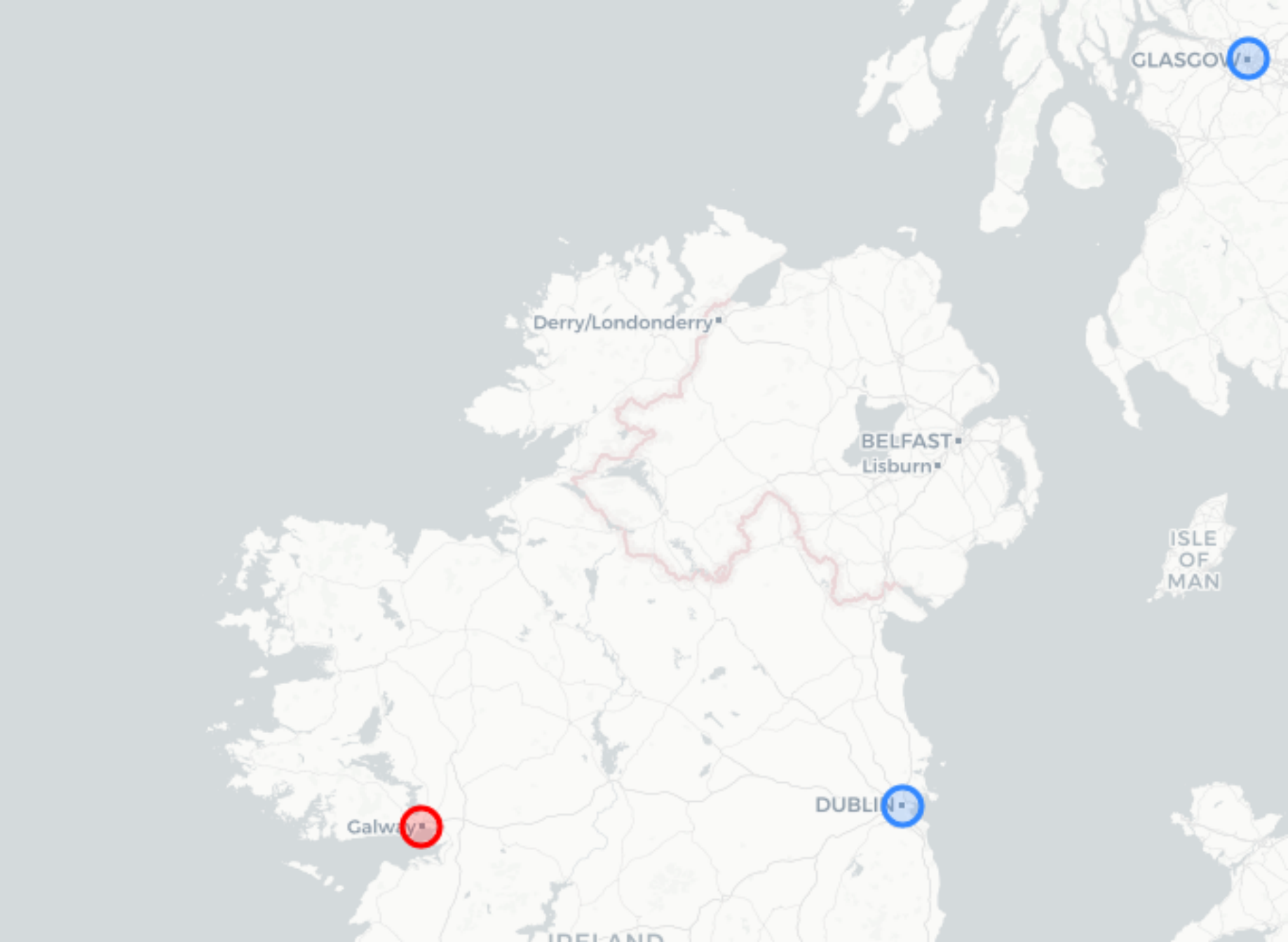
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Galway (and BAIL)

Regional Ocean Modeling System (ROMS)

ROMS is a free-surface, terrain-following, primitive equations ocean model; see [?, ?]. Also .

It is rather complex... some (numerical analysis) highlights

- 2D and 3D equations are time-discretized using a third-order accurate predictor (Leap-Frog) and corrector (Adams-Molton) time-stepping algorithm;
- The default stencil uses centered, second-order finite differences on a staggered vertical grid.
- In the horizontal, the primitive equations are evaluated using boundary-fitted, orthogonal curvilinear coordinates on a staggered Arakawa C-grid.

[Need coarser grid]

Figure: Salinity in Galway Bay, predicted by ROMS

Issue 1: The split-explicit time stepping procedure for an oceanic model (e.g. ROMS, [?]) implies that the barotropic mode system (vertical integrals of horizontal velocities, coupled with contribution of free-surface elevation into vertical integrals of pressure-gradient terms) is solved separately from the rest of the model using **smaller time step**.
This leads to significant computational savings, because the **large number of short time steps are applicable only for 2D part** of the whole 3D model.
However, this also results in more complicated code, carefully designed to avoid numerical errors and instability.

Issue 2: At the same time, vertical processes—mixing of tracers, viscous exchange of momenta, and, recently, vertical advection (only where it is strictly unavoidable: [**?**])–are treated implicitly, but only in a one-dimensional manner resulting in a simple and efficient solver.

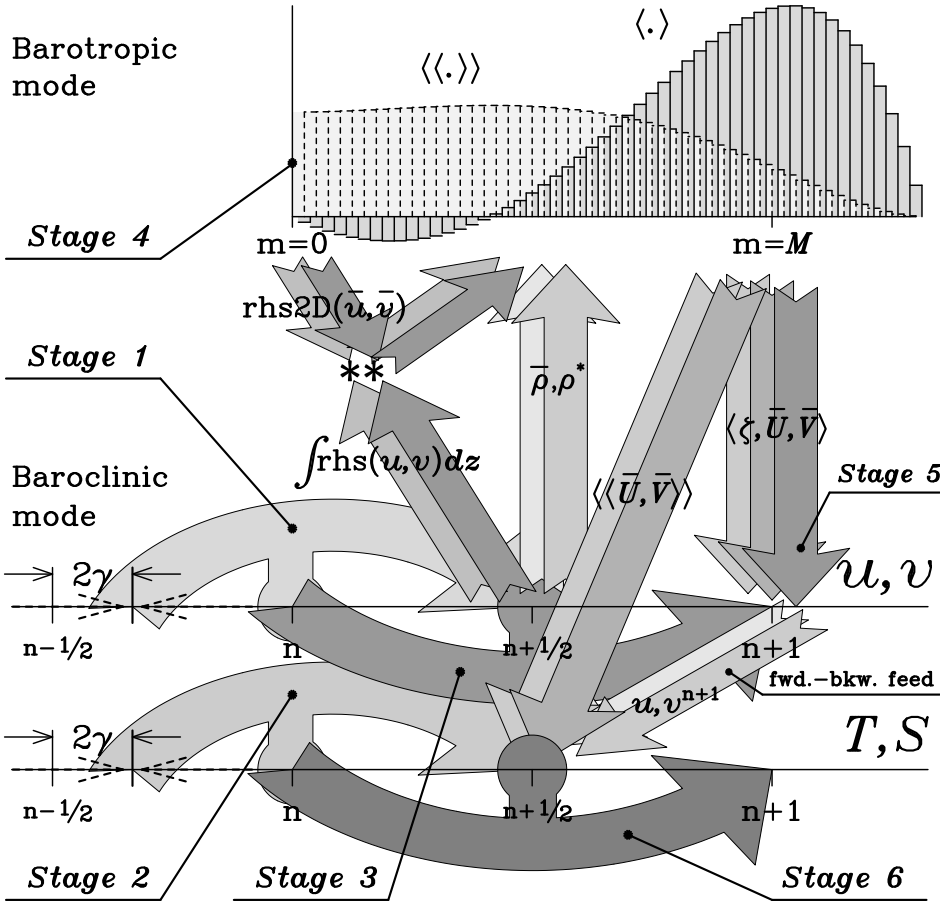
Issue 3: Also require parameterization of vertical profile of turbulent mixing coefficient along with kinematic stress bottom boundary condition, which is of the no-slip type, but nonlinear in nature due to the fact that both bottom drag coefficient and vertical viscosity profile depend on the magnitude of the current ([**?**]).

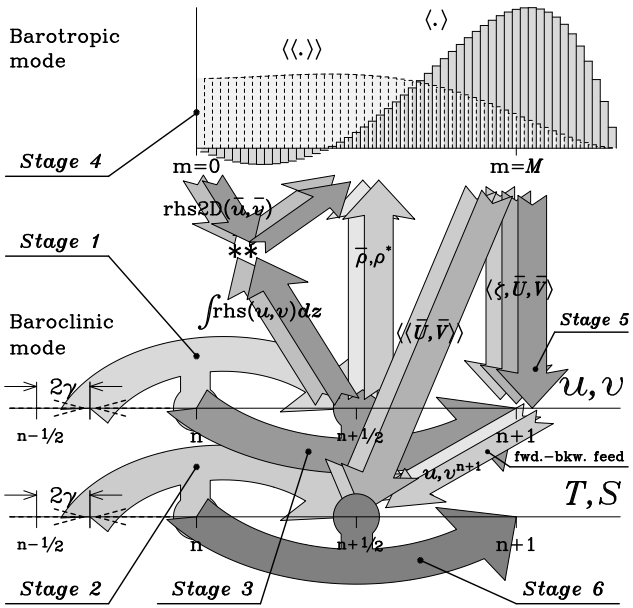
Problem: the barotropic mode needs to know the bottom drag terms in advance (which can be computed only within the 3D part of the code), but when done, the result barotropic mode calculation adjusts the horizontal velocity components in the 3D mode, compromising both the no-slip boundary conditions and the consistency of bottom stress with vertical viscosity profile.

The plan

In this study we propose modification to existing algorithms to eliminate the splitting errors associated with bottom drag.
Special attention is also paid to a **bottom boundary condition**, capable of dealing with under-resolved boundary layer in vertical mixing parameterization scheme.

ROMS time stepping algorithm with barotropic mode splitting and coupling during corrector stage





- Arched horizontal arrows indicate progression of slow-time (baroclinic) variables: u, v, T, S.
- Circles indicate computation of r.h.s. for the respective variables, u, v, T, S.
- 4 ascending arrows represent computation of 3D→BM forcing terms: vertically integrated r.h.s. terms for 3D momentum equations and 2-way vertically averaged densities, $\overline{\rho}$ and ρ_*.
- 5 descending arrows represent updated barotropic barotropic fluxes (vertically integrated velocities), which are 2-way averaged in fast time.
- Vertical integrals of inermediate values $u^{n+1/2}, v^{n+1/2}$ (computed during predictor stage) are forced the respective barotropic values at the previous time step n.

Why bottom drag term does not naturally fit into this data flow and requires special attention?

The **bottom drag** term relates to the net force applied to the fluid in bottom-most grid box $\Delta z_1 \equiv \Delta z_{k=1}$ with velocity inside it $u_1 = u_{k=1}$,

$$\frac{\partial}{\partial t}(\Delta z_1 \cdot u_1) + \cdots ??? = -r_D \cdot u_1$$

thus is to be interpreted in finite-volume sense.

The **drag coefficient**, r_D, may be constant, i.e., a finite-difference or finite-volume approximation based on constant viscosity, no-slip boundary condition and assumption about velocity profile (hence its derivative at the bottom), or it may depend on velocity itself (nonlinear, as it happens in turbulent mixing schemes).

The situation when $r_D \cdot \Delta t/\Delta z_1 > 1$, where Δt is the baroclinic time step, requires impicit treatment of vertical viscosity, including bottom b.c.: it attempts to remove more momentum from grid-box Δz_1 than is available there, leading to oscillations and instability.

During the coupling stage, barotropic-mode corrections to the 3D-mode u and v components are applied uniformly thoughout the entire vertical column, which unavoidably alters the bottom-most u_1, v_1, thus distorting the no-slip bottom b.c.

Recomputing bottom drag term after the barotropic corrections are applied restores the no-slip bottom b.c., but distorts the state of approximate nondivergence of barotropic fluxes.

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